

# AI Newsletter Automation — Framework Document

AbaKus, The Tech Club of IIM Kozhikode — 2026-27 Selection Task

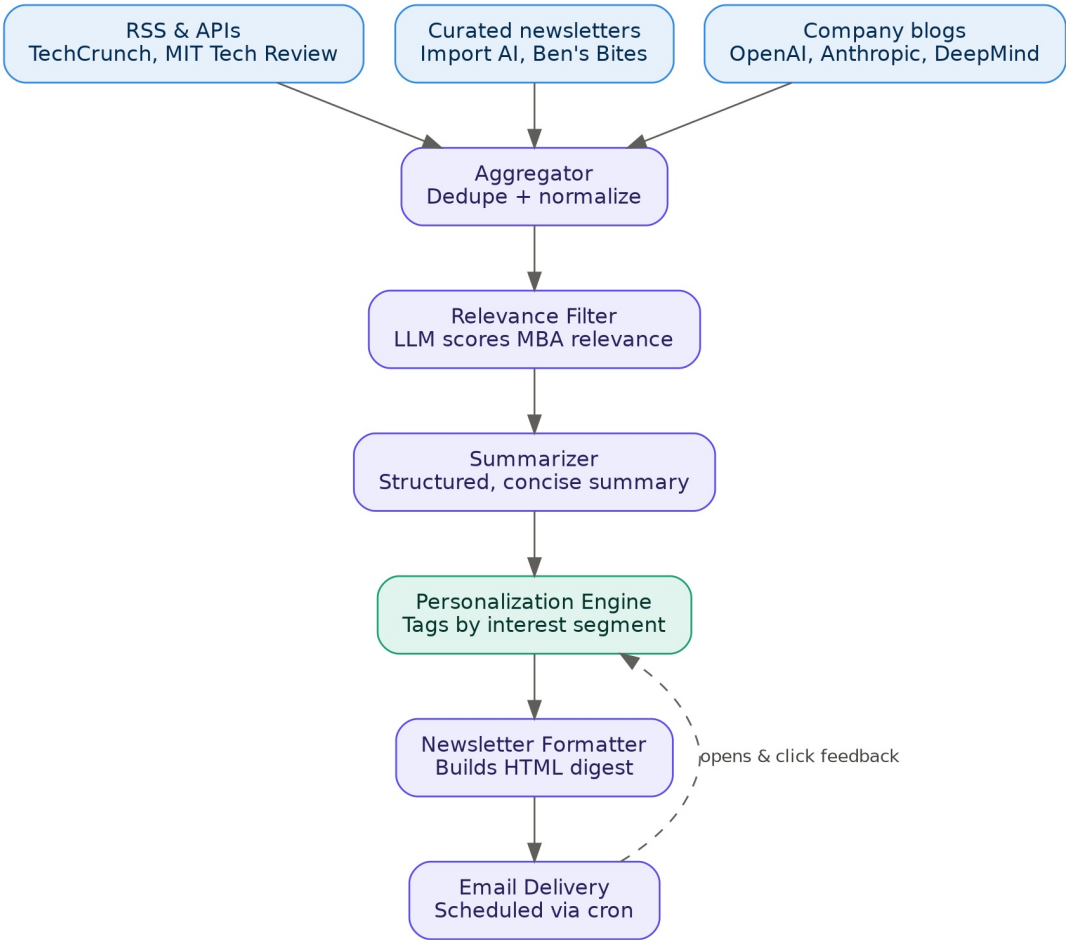
## 1. Problem Framing

MBA students don't lack access to AI news — they're drowning in it. The real gaps are:

| Problem            | Why it matters                             | How this system addresses it                                |
|--------------------|--------------------------------------------|-------------------------------------------------------------|
| Signal vs. noise   | Most "AI news" has zero business relevance | LLM-based relevance scoring, not keyword matching           |
| Translation gap    | Technical news ≠ business insight          | Every item gets a "Why MBA students should care" line       |
| Engagement fatigue | Same format every week → unsubscribes      | Rotating feature sections + feedback-driven personalization |

**Design principle:** optimize for *relevance per minute read*, not volume of news.

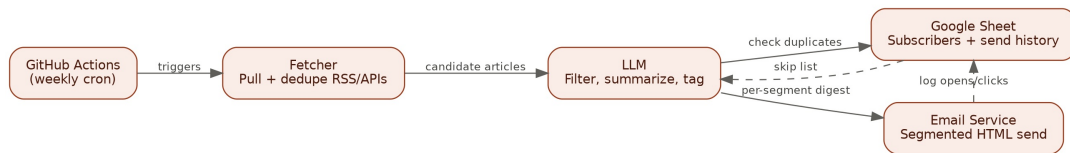
## 2. System Architecture



System architecture

**Why this shape:** a linear pipeline keeps each stage independently testable and replaceable (e.g., swap the LLM provider, add a Telegram formatter, add a new source) without touching the rest.

## 3. Weekly Automation Workflow



Weekly automation workflow

Runs unattended once scheduled — no server to maintain, no manual curation.

## 4. Content Selection Logic

Every article is scored 0–10 on four axes before it qualifies for the newsletter:

1. **Business/strategic impact** — does this change how companies compete or operate?
2. **Career relevance** — is this a tool/trend recruiters or case interviews will reference?
3. **Case-study potential** — could this become a discussion point or CV line?
4. **Novelty vs. noise** — is this genuinely new, not a rehash?

Only items above a composite threshold survive. This rubric — not a keyword list — is what drives content quality.

## 5. Personalization Logic

- One-time signup form captures interest tags: Finance / Consulting / Marketing / Product / Analytics
- Every summarized article is tagged with the same categories
- Formatter filters per-subscriber before send — simple, explainable, no ML recommender needed at this scale
- Feedback loop (thumbs-up/thumbs-down per section, tracked via UTM links) reweights future scoring per user over time

## 6. Tech Stack (cost-efficient by design)

| Layer      | Choice                     | Why                                          |
|------------|----------------------------|----------------------------------------------|
| Sources    | RSS + free APIs            | No paid data feeds needed                    |
| Processing | Python script              | Simple, portable, easy to extend             |
| LLM        | Claude API (batched calls) | One call per run, cached by article URL hash |
| Storage    | Google Sheets              | Zero-infra subscriber DB + send history      |
| Scheduling | GitHub Actions cron        | Free, versioned, reliable                    |
| Delivery   | Gmail API / Brevo / Resend | Free tier covers small subscriber base       |
| Optional   | Twilio WhatsApp API        | For a 3-line digest to non-email users       |

## 7. Differentiator Features (prioritized)

| Feature                                 | Effort | Value                               |
|-----------------------------------------|--------|-------------------------------------|
| “Why MBA Students Should Care”          | Low    | High — core to relevance            |
| AI Tool of the Week (with MBA use case) | Low    | High — practical takeaway           |
| Feedback-based personalization          | Medium | High — retention driver             |
| WhatsApp digest                         | Medium | Medium — reach beyond email         |
| Infographics/quizzes                    | High   | Low-Medium — polish, not core value |

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## 8. Evaluation Criteria Mapping

| Criteria                     | Weight | Addressed by                                                |
|------------------------------|--------|-------------------------------------------------------------|
| Functionality & Reliability  | 30%    | End-to-end pipeline, cron automation, dedup logic           |
| Content Selection Quality    | 20%    | 4-axis LLM relevance rubric                                 |
| Automation & Scalability     | 20%    | GitHub Actions + stateless pipeline + Sheets DB             |
| Code Quality & Documentation | 10%    | This document + modular script design                       |
| Creativity & Value Addition  | 20%    | MBA-specific framing, feedback loop, multi-channel delivery |

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## 9. Build Sequence

1. Source fetcher + relevance filter script
2. Summarizer + newsletter formatter
3. GitHub Actions automation
4. Sample newsletter output
5. Final documentation polish